1. In a network, large application-layer messages are divided into smaller units called packets before being transmitted. For example, a large PDF file is segmented into smaller packets, which are then sent across the network. The receiving computer reassembles these packets back into the original document. The process of splitting messages into packets is called message segmentation.

   Consider the following scenario:

   1. You need to send a file that is 800,000 bits in size from a server to a client.
   2. The network consists of three links: from the server to Router 1 (R1), from R1 to Router 2 (R2), and from R2 to the client.
   3. Each link in the network has a transmission rate of 4 Mbps (megabits per second).
   4. The propagation, queuing, and processing delays are negligible.

   The network topology is illustrated in Fig. 1.

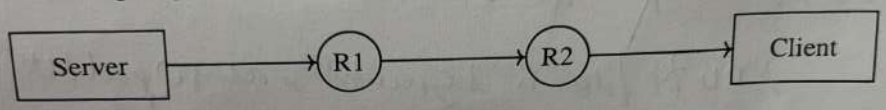


Figure 1: Network topology with server, two routers, and client

[3] (a) Considering that each router operates in store-and-forward mode, calculate the total time required to transmit the file from the server to the client without message segmentation?

Transmission rate = 4×10^6 bits per second.

Since propagation, queuing & processing delays are negligible.

Time to transmit the file from one link to another = Total size (in bits) / Transmission rate (bits/s)

$= \frac{8 \times 10^5}{4 \times 10^6}$ seconds $= 0.2$ seconds

Time to transmit file from server to client without message segmentation = 0.2 seconds

Total time taken to transmit the file = no. of links × time taken in one link

$= 3 \times 0.2 = 0.6$ seconds

3

3 (b) Now suppose that the file is segmented into 80 packets, with each packet being 10,000 bits long. Calculate the total time required to transmit the file from the server to the client when the file is segmented?

Since the file is very segmented.

It will first ~~so from server~~ to R_1 & get reassembled.

then R_1 to R_2

~~the R_2 to client~~

Time ~~to transmit from~~ from server to

since the file is segmented, all 80 packets are sent together.

Time to transmit from one link to another = $\frac{0.2 \text{ seconds}}{800}$

$= \frac{1 \times 10^{-1}}{4 \times 10^{2}}$ seconds

$= \frac{1}{400}$

$= \frac{0.25}{100}$ seconds

$= 0.0025$ seconds

Time to transmit the file = 2×0.025 seconds

$= 0.075$ seconds

0

2. A periodic signal of the form shown in Fig. 2 has a Fourier series expansion given by

$$m(t) = \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos\left(\frac{2\pi(2n-1)t}{T}\right)$$

6.5 ~~10.5~~

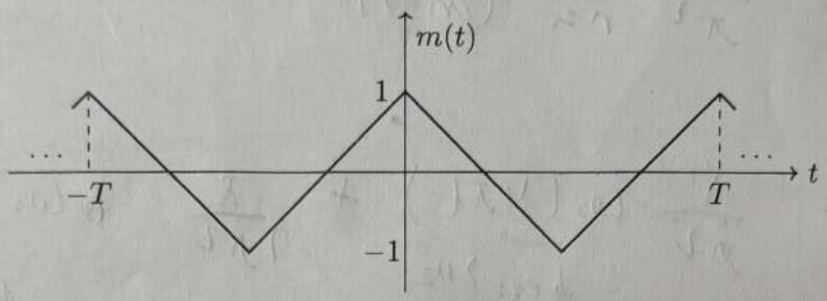


Figure 2: Plot of $m(t)$

2 (a) Select T such that the fundamental frequency of $m(t)$ is 2 Hz.

~~frequency of m(t)~~

fundamental frequency = frequency at $n=1$ = $\frac{1}{T} = 2\text{Hz}$

$T = \frac{1}{2}\ \text{sec} = 0.5\ \text{sec}$.

2

2 (b) Suppose that $m(t)$ is passed through a low pass filter with cut-off frequency of 8 Hz to yield $\tilde{m}(t)$. Write an expression for $\tilde{m}(t)$.

$$m(t) = \frac{8}{\pi^2} \sum_{n=2}^{\infty} \frac{1}{(2n-1)^2} \cos(4\pi(2n-1)t) \quad (T=\tfrac{1}{2})$$

$$\tilde{m}(t) = \frac{1}{\pi^2}\cos(4\pi t) + \frac{8}{9\pi^2}\cos(12\pi t)$$

$f_{c4} = 2\,\text{Hz}$ $\qquad f_{c4} = 3\,\text{Hz}$.

2

3 (c) Given that the power of $m(t)$ is $P_m = 1/3$ and power of $\tilde{m}(t)$ if $P_{\tilde{m}}$, calculate the percentage of power is lost due to the filtering operation, i.e., $\left(1 - \frac{P_{\tilde{m}}}{P_m}\right) \times 100\%$.
If you do not have a calculator, you may leave your answer in the form of an expression.

$\tilde{m}(t) = \frac{8}{\pi 2} \cos(4\pi t) + \frac{8}{9\pi 2} \cos(12\pi t)$

For periodic signal $P_{\tilde{m}} = \frac{1}{T_0} \int_0^{T_0} |A_m \cos(2\pi f_0 t + \theta)|^2 dt$.

A $A\cos(2\pi f_0 t + \theta)$

$P_m = \frac{A_m^2}{2}$.

$P_{\tilde{m}} = \frac{\left(\frac{8}{\pi 2}\right)^2}{2} + \frac{\left(\frac{8}{9\pi 2}\right)^2}{2}$.

$\left(1 - \frac{P_{\tilde{m}}}{P_m}\right) \times 100\% = \left(1 - \frac{1}{2}\left(\frac{8}{\pi 2}\right)^2 \left(\frac{1 + \frac{1}{9^2}}{\frac{1}{3}}\right)\right) \times 100\%$.

$= \left(1 - \frac{32}{\pi^4} \times \left(3 + \frac{1}{27}\right)\right) \times 100\%$

$= \left(1 - \frac{32}{\pi^4} \times \frac{82}{81}\right) \times 100\%$.

∴ power lost due to filtering operation.

2.5 + 0.5 = 3

3. Design a double sideband (DSB) modulator to generate the signal

$$x_c(t) = Am(t)\cos(2\pi f_c t + \theta)$$

where the baseband message signal $m(t)$ has bandwidth B and A may be any positive number and $\theta \in [0, 2\pi)$. You must use the modulator available in the stock room, shown in Fig. 3 , where the carrier generator available generates $\cos^3(2\pi f_c t)$ (instead of $\cos(2\pi f_c t)$).

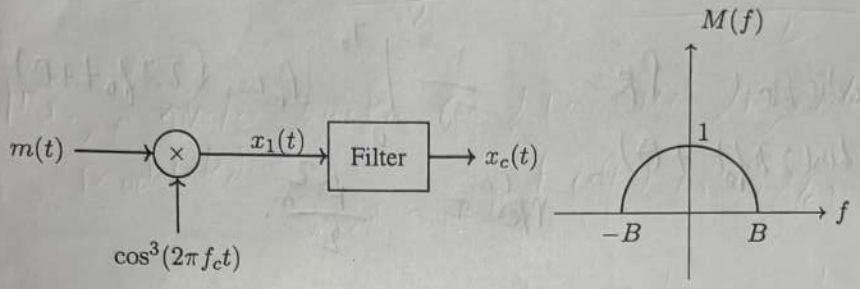
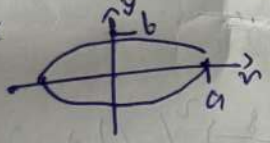


Figure 3: DSB Modulator and $M(f)$

3 (a) Plot the spectrum of $x_1(t)$.



$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$

$|M(f)| = \sqrt{1 - \frac{f^2}{B^2}}$

$\cos 3\theta = 4\cos^3\theta - 3\cos\theta.$

$x_1(t) = m(t)\cos^3(2\pi f_c t) = \frac{1}{4}m(t)\left(\cos(6\pi f_c t) + 3\cos(2\pi f_c t)\right)$

$x_1(t) = \frac{m(t)}{2}\left(e^{j6\pi f_c t} + e^{-j6\pi f_c t} + 3e^{j2\pi f_c t} + 3e^{-j2\pi f_c t}\right)$

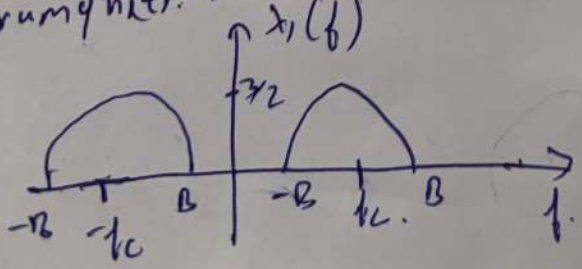
$X_1(f) = \frac{M(f-3f_c)}{2} + \frac{M(f+3f_c)}{2} + 3\frac{M(f-f_c)}{2} + 3\frac{M(f+f_c)}{2}$

since $m_c(t) = Am(t)\cos(2\pi f_c t + \theta) \Rightarrow |B| < 3f_c$

$B = f_c$. $M(f-3f_c) = M(f+3f_c) = 0$

0.5

Spectrum of $x_1(t)$:-



2 (b) For the filter, the following options are available. Which one of these should be used and why?

A. A low pass filter with cut-off frequency B.
B. A low pass filter with cut-off frequency $2B$.
C. A band pass filter centered at f_c with bandwidth B.
D. A band pass filter centered at f_c with bandwidth $2B$.

A bad pass filter centred at fc should be chosen since we need to produce a signal of frequency fc
A low pass filter is used when we need to separate. m(t) from m1(t) which is not the case here.
Bandwidth should be B since spectrum of s(t) should not overlap.

2 (c) Can a similar system (with a possibly different filter) still be used to generate DSB-SC modulated signal if the carrier generator generates $\cos^2(2\pi f_c t)$?

$n_1(t) = m(t)\cos^2(2\pi f_c t)$

$n_1(t) = \frac{m(t)}{2} + \frac{m(t)\cos(4\pi f_c t)}{2}$

No, a similar system can't be used to generate DSB-SC modulated signal $n(t) = Am(t)\cos(2\pi f_c t + \theta)$

As we will only be able to generate signal of frequency $2f_c$.

We want to generate $Am(t)\cos(2\pi f_c t + \theta)$

2